Warm Starts, Honest Clocks: Dual-Horizon Radar Alignment for Distributed Replay

***Abstract*—Distributed automotive radar replay must initialize calibration state as well as object tracks. We formulate an eligibility-indexed dual-horizon alignment schedule and distinguish causal cold-start adaptation from provenance-qualified historical seeding. Exact bias and variance expressions expose a finite startup-gain budget, while yaw-invariant range–rate relations separate timestamp offset from radial-speed bias. Reproducing a ten-seed scalar experiment gives 46.2 frames to first tolerance entry; a 100-seed extension averages 41.14 frames. Two alternative gain settings remain censored at 400 frames under the same protocol, demonstrating that a fast-lock claim cannot be transferred between configurations. Controlled timing fixtures attain 6.632 and 3.799 ms RMSE in radial and tangential geometries. The study provides a reproducible initialization argument rather than inferring cluster throughput from estimator delay.**

***Index Terms*—automotive radar, dynamic alignment, distributed replay, timestamp calibration**

I. Introduction

The first vehicle in a replay shard is rarely the first vehicle the original radar saw. By the time a continuous drive reaches a shard boundary, the sensor has already learned a mounting correction, accumulated clutter statistics, and established tracks. A worker starting from an empty state inherits none of that history. The resulting error can look like a tracking failure even when it originates much earlier, in the calibration that rotates every detection entering the tracker. This distinction matters when a long drive is divided among compute workers. A short preceding segment may be sufficient to initialize position and velocity, yet insufficient to recover a calibration state that updates only during suitable motion. Adding the same overlap to every worker is therefore a convenient scheduling rule, not a convergence argument. The relevant horizon is the number and quality of informative updates. We study two complementary answers. During a genuine cold start, an eligibility-indexed gain allocates more adaptation to the beginning of the sequence and then returns to a conservative steady regime. During historical replay, a valid recorded state can initialize the worker directly. These answers must remain separate: a retrospective state is useful for reproducing a recording, but it would leak future information into an online cold-start experiment. The paper makes three contributions. First, it derives a calibration/replay contract that distinguishes elapsed frames, accepted updates and state provenance. Second, it provides finite-horizon error and noise analysis for a dual-horizon recursion, together with a yaw-independent clock-bias formulation for aligning asynchronous observations. Third, it revisits the experimental claims under explicit configurations. Reproducing the original ten-seed race yields 46.2 frames to first tolerance entry for one gain setting; applying two other, previously conflated settings does not reach that tolerance within the same 400-frame experiment. The difference is central to the paper: fast initialization is a property of a complete estimator configuration, not a formula printed beside a desirable number.

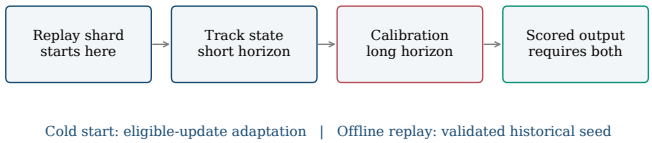


Fig. 1. Replay initialization has two horizons. Scoring is valid only after the state required by the entire perception path is available. Historical seeding and cold-start adaptation solve different operating cases.

II. Related Work and Positioning

Recursive state estimation provides the mathematical foundation [1]; it does not prescribe a universal warm-up duration. Calibration adds a second source of state whose observability depends on scene geometry and vehicle motion. Our concern is the interface between this state and distributed replay, rather than a claim that exponential gain schedules are new in isolation. Temporal parallelization of Bayesian filtering and smoothing replaces sequential recursions by associative operators under suitable models [7]. That addresses computational dependency inside an estimator. It does not automatically restore the state of a nonlinear production front end, nor preserve decisions made by thresholded association logic. We therefore treat replay initialization and temporal parallel filtering as complementary problems. The timing analysis uses rotation-invariant inner products to separate clock offset from yaw under a stated kinematic model. The residual beat estimator is an application of phase-increment frequency estimation, whose established lineage includes Kay [3]. We include it as a resource-aware supporting component, not as the central novelty of calibration initialization. The distinction keeps the main argument focused: replay should either reconstruct a valid state or establish why its initialization error cannot change the scored decisions.

III. Problem Formulation

Let $\theta^\star$ be a locally constant boresight angle and $\hat{\theta}_k$ its estimate after elapsed frame k. A binary eligibility variable a_k

indicates whether that frame supplies a trustworthy calibration observation. The accepted-update counter is

$$N_k = \sum_{i=0}^{k} a_i. \tag{1}$$

Let $z_k = \theta^\star + \eta_k$ denote a local angle observation. An explicit two-regime recursion is

$$\hat{\theta}_{k+1} = \hat{\theta}_k + g_k(z_k - \hat{\theta}_k), \tag{2}$$

$$g_k = K_s + a_k K_f \exp(-\alpha N_k). \tag{3}$$

The gains K_s, K_f are dimensionless and α is per accepted update. When $a_k = 0$, this model still applies the slow update. A strict rejection policy would instead set $g_k = 0$; that is a different estimator and must not be substituted silently in a comparison. The observation model is a local reduction of a calibration problem, not an identification of an inaccessible production filter. It is useful because it exposes the effect of the startup schedule without hiding it behind a large software stack. The analysis assumes a fixed angle over the evaluated horizon, a documented sensor/ego coordinate convention, and a declared policy for rejected observations.

A. What does "lock" mean?

We distinguish first tolerance entry,

$$T_{\text{first}} = \inf\{k+1 : |\hat{\theta}_k - \theta^\star| < \delta\}, \tag{4}$$

from persistent lock over H successive frames. First entry is easier to achieve under noise and says nothing about a later departure. An estimate-change test, such as $|\hat{\theta}_k - \hat{\theta}_{k-1}| < \delta$, measures motion of the estimator rather than error relative to the true mounting angle. A frozen but wrong estimate passes that test. All experimental lock results below refer to first entry unless explicitly stated otherwise.

IV. Dual-Horizon Analysis

A. Bias contraction

With $e_k = \hat{\theta}_k - \theta^\star$, the noiseless error satisfies

$$e_m = e_0 \prod_{k=0}^{m-1} (1 - g_k). \tag{5}$$

For $0 < g_k < 1$, its magnitude contracts monotonically. Under independent zero-mean observation noise, the same product governs the mean error. In the special case of exclusively eligible observations, $N_k = k + 1$ and

$$\frac{|e_m|}{|e_0|} \le \exp\left[-mK_s - K_f \frac{e^{-\alpha}(1 - e^{-m\alpha})}{1 - e^{-\alpha}}\right]. \tag{6}$$

This follows from $\log(1 - u) \le -u$. It provides a sufficient contraction bound; it is not an exact lock-time expression in the presence of noise. The finite sum also explains a subtle limitation. The fast component supplies a finite total adaptation budget, whereas the slow component accumulates indefinitely. If the initial fast budget is too small, a schedule can decay almost completely before the estimate reaches the desired tolerance. The final stage then reverts to the same slow convergence one was trying to avoid.

B. Noise cost of moving quickly

If the eligibility sequence is fixed and η_k is independent with variance σ_k^2, then

$$P_{k+1} = (1 - g_k)^2 P_k + g_k^2 \sigma_k^2. \tag{7}$$

The exact finite-horizon expression separates transported initialization uncertainty from injected measurement uncertainty:

$$P_m = P_0 \prod_{i=0}^{m-1} (1 - g_i)^2 + \sum_{j=0}^{m-1} g_j^2 \sigma_j^2 \prod_{i=j+1}^{m-1} (1 - g_i)^2. \tag{8}$$

For a fixed slow gain and constant variance, $P_\infty = K_s \sigma^2 / (2 - K_s)$. A larger startup gain suppresses initialization bias but admits more early measurement noise. The stationary-scatterer gate is therefore part of the estimator, not a decorative safeguard added after tuning.

C. Physical qualification

For a stationary world point, let $\mathbf{u}$ be the sensor-to-point unit vector and $\mathbf{v}_s$ the sensor velocity in the same frame. A Doppler consistency test is

$$|v_r + \mathbf{u}^T \mathbf{v}_s| \le \epsilon_v. \tag{9}$$

Lever-arm rotation contributes to $\mathbf{v}_s$ and cannot be ignored on a turning vehicle. A Doppler alias must be resolved before applying this test. The gate admits stationary guardrails when they satisfy the model; it does not reject them merely because they are infrastructure. Independent geometric diversity and robust angle fitting are still needed to prevent a poorly observed scene from producing a confident mounting correction.

V. Preflight Seeding Without Look-Ahead Confusion

Historical replay offers information that an online cold start does not. A recorded alignment state can be attached to a worker's initial condition, provided its meaning is explicit. Denote the seed by $(\theta_0, P_0, \mathcal{P})$, where $\mathcal{P}$ records sensor identity, mounting configuration, epoch and quality. Acceptance requires compatibility with the replayed sensor and a documented relationship between the state epoch and the scored segment. A mounting-angle range check catches gross corruption but does not establish convergence. Finite-value checks, age, configuration consistency and uncertainty must be considered separately. If only a point estimate is available, treating it as exact silently understates uncertainty. Where a covariance is available, the seeded error is propagated using (7) rather than reset to zero by assertion. An extracted example in the project records azimuth $-1.145°$. That is a concrete transport example, not an accuracy guarantee for every worker. The HDF extraction prototype averages a tail window and uses record length as a convergence proxy; a production criterion should consume the recorded quality and uncertainty channels directly. Offline reconstruction and online evaluation must use

TABLE I
ALIGNMENT CONFIGURATIONS; FIRST-ENTRY HORIZON 400 FRAMES

Setting	K_f	K_s	α	Censored
F1	0.15	0.004	0.05	0/100
F2	0.05	0.0025	0.08	100/100
F3	0.05	0.0025	0.1	100/100

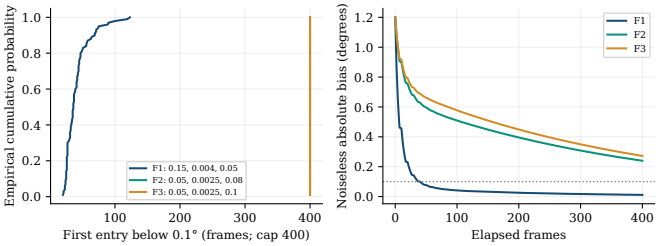


Fig. 2. Configuration-controlled alignment study. Left: first-entry distributions over 100 seeds; a mass at 400 denotes censoring. Right: exact noiseless bias for the same eligibility pattern. F1 uses $(K_f, K_s, \alpha) = (0.15, 0.004, 0.05)$; F2/F3 use $(0.05, 0.0025, 0.08/0.10)$.

distinct modes so a future-converged value cannot inadvertently improve a claimed online cold-start result.

VI. TEMPORAL ALIGNMENT OF ASYNCHRONOUS SENSORS

Calibration consistency is incomplete if sensors are fused at different epochs. Under the wrapper's stated ego-frame convention, a second-order correction is

$$\mathbf{p}_c = R(\omega\Delta t)\mathbf{p} - \mathbf{v}_e\Delta t - \tfrac{1}{2}\mathbf{a}_e\Delta t^2. \tag{10}$$

Position, velocity, acceleration and yaw rate use metres, m/s, m/s^2 and rad/s, respectively. The sign of Δt and the direction of the frame map define the rotation sign; the formula is not transferable between coordinate conventions without that definition.

A. Clock offset before yaw

Let an independent track predict $\mathbf{d}$ and velocity $\mathbf{v}$ at a reference epoch. For a constant-velocity offset τ, the contemporaneous point is $\mathbf{x} = \mathbf{d} + \tau\mathbf{v}$. Its range and radial velocity satisfy

$$rw - \mathbf{d}^T\mathbf{v} = \|\mathbf{v}\|^2\tau. \tag{11}$$

A common yaw rotation does not change the inner product. With additive radial-speed bias b, define $y_i = r_i w_i - \mathbf{d}_i^T\mathbf{v}_i$ and $a_i = \|\mathbf{v}_i\|^2$. Two targets give

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} a_1 & r_1 \\ a_2 & r_2 \end{bmatrix} \begin{bmatrix} \tau \\ b \end{bmatrix}. \tag{12}$$

The determinant $a_1 r_2 - a_2 r_1$ controls identifiability. A tangential target is not automatically singular: nonzero speed still supplies timing information. Once time is corrected, yaw can be estimated from a weighted circular resultant of predicted and observed bearings. This ordering avoids asking a single optimizer to explain the same discrepancy simultaneously as time, yaw and radial-speed bias.

VII. EXPERIMENTAL EVALUATION

A. Alignment protocol

The scalar race starts at 0° with a true offset of 1.2°. Each ten-frame block contains seven eligible frames with Gaussian observation standard deviation 0.3° and three other frames with 1.5° standard deviation. Both methods receive the same observations. We compare the original ten seeds and a 100-seed extension. The threshold is 0.1° and the evaluation ends at frame 400. A sequence that never enters the threshold is right-censored, not assigned a measured lock time of 400 frames. The original F1 ten-seed result is a mean first entry of 46.2 frames. Its 100-seed mean is 41.14 frames, with no censored fast sequences; all slow sequences remain censored. F2 and F3 remain censored in all 100 trials under this observation protocol. Therefore the manuscript does not attach "46 frames" to their parameters. Nor does it interpret a censored 400-frame baseline as a known 400-frame convergence time. At an assumed 20 Hz, the original mean corresponds to 2.31 s. File counts, recording duration and compute runtime are different quantities. No twenty-node wall-clock speedup or 360,000-frame production loss is inferred from this scalar experiment.

B. Timing and supporting estimators

The clock fixture recovers $\tau = 20$ ms and $b = 0.1$ m/s to numerical precision for its exact constant-velocity model. In 2,000 noisy draws per geometry, the single-target timing RMSE is 6.632 ms for the radial fixture and 3.799 ms for the tangential fixture. Position and velocity component noise have standard deviation 0.01 in their respective units; range noise is 0.01 m and radial-speed noise is 0.005 m/s. These controlled fixtures test the invariant, not a production synchronization service.

C. Resource-aware residual estimation

For a single complex FMCW tone, derotating a predicted beat frequency leaves a near-DC residual whose lag-one phase estimates frequency [3]. The available prototype uses 512 samples at 10 MHz, a 77 GHz carrier and chirp slope 5×10^{12} Hz/s. Its operation-count model gives 4,608 versus 1,024 complex products, a 4.5 ratio, but does not time an optimized DSP FFT. The recorded errors 0.4393, 0.1059 and 0.1075 m are mean absolute range error, despite an older print label calling them RMSE. The FFT interpolation implementation also requires a sign-correct denominator before serving as a strong accuracy baseline. We consequently use this study to motivate the compute envelope, not to claim superiority over a competitive spectral estimator.

VIII. DISCUSSION AND CONCLUSION

The practical lesson is to stop treating replay overlap as a substitute for an initialization model. Fast adaptation can reduce the first-entry delay, but only for a qualified observation stream and a gain budget large enough to remove the initial bias. Historical seeding can remove that adaptation phase in offline reconstruction, but only by making the provenance

and evaluation mode explicit. The present evidence supports a controlled algorithm study, exact clock fixtures and an inspectable integration design. The production wrapper's gain readout and update-count cap are not a replacement for the buried calibration recursion. Persistent-lock tests, quality-qualified header extraction and a complete production replay comparison remain the next system-level evaluation. Those boundaries do not diminish the central result; they make clear which part of the replay problem has actually been solved.

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